Worksheet 10: SERT

Introduction

In the Spatio-temporal Encoder Representations from Transformers (SERT) architecture, we do not treat weather data as a rigid grid. Instead, we treat observations as a sequence of tuples. For this example, we define our input data as $x = (t, l, f, v)$ where:

- t: Time index
- l: 1D Location (a simplified coordinate)
- f: Feature index (e.g., 1 for Temperature, 2 for Pressure)
- v: Normalized physical value

Step 1: The Input Data

Assume we have two raw weather observations. We represent them as column vectors:

$$x_1 = \begin{pmatrix} t_1 \\ l_1 \\ f_1 \\ v_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0.5 \end{pmatrix}, \quad x_2 = \begin{pmatrix} t_2 \\ l_2 \\ f_2 \\ v_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 2 \\ 0.4 \end{pmatrix}$$

Step 2: Component Embeddings

Instead of embedding the entire tuple at once, the SERT method applies a simple 1-hidden layer Neural Network (simplified here to a linear projection without bias) to compute an embedding of dimension 2 for each individual component.

We are given the following weight vectors for each component:

$$W_t = \begin{pmatrix} 0.1 \\ 0.2 \end{pmatrix}, \quad W_l = \begin{pmatrix} 0.2 \\ 0.1 \end{pmatrix}, \quad W_f = \begin{pmatrix} 0.1 \\ 0.1 \end{pmatrix}, \quad W_v = \begin{pmatrix} 1.0 \\ 2.0 \end{pmatrix}$$

Let us calculate the embeddings for the first observation, x_1:

$$e_{t1} = W_t \times t_1 = \begin{pmatrix} 0.1 \\ 0.2 \end{pmatrix} \times 1 = \begin{pmatrix} 0.1 \\ 0.2 \end{pmatrix}$$

$$e_{l1} = W_l \times l_1 = \begin{pmatrix} 0.2 \\ 0.1 \end{pmatrix} \times 2 = \begin{pmatrix} 0.4 \\ 0.2 \end{pmatrix}$$

$$e_{f1} = W_f \times f_1 = \begin{pmatrix} 0.1 \\ 0.1 \end{pmatrix} \times 1 = \begin{pmatrix} 0.1 \\ 0.1 \end{pmatrix}$$

$$e_{v1} = W_v \times v_1 = \begin{pmatrix} 1.0 \\ 2.0 \end{pmatrix} \times 0.5 = \begin{pmatrix} 0.5 \\ 1.0 \end{pmatrix}$$

Next, we calculate the embeddings for the second observation, x_2:

$$e_{t2} = W_t \times t_2 = \begin{pmatrix} 0.1 \\ 0.2 \end{pmatrix} \times 2 = \begin{pmatrix} 0.2 \\ 0.4 \end{pmatrix}$$
$$e_{l2} = W_l \times l_2 = \begin{pmatrix} 0.2 \\ 0.1 \end{pmatrix} \times 3 = \begin{pmatrix} 0.6 \\ 0.3 \end{pmatrix}$$
$$e_{f2} = W_f \times f_2 = \begin{pmatrix} 0.1 \\ 0.1 \end{pmatrix} \times 2 = \begin{pmatrix} 0.2 \\ 0.2 \end{pmatrix}$$
$$e_{v2} = W_v \times v_2 = \begin{pmatrix} 1.0 \\ 2.0 \end{pmatrix} \times 0.4 = \begin{pmatrix} 0.4 \\ 0.8 \end{pmatrix}$$

Step 3: Summing the Embeddings

To create the final latent representation E_i for each observation, we calculate the sum of the embedded values.

For x_1:

$$E_1 = e_{t1} + e_{l1} + e_{f1} + e_{v1} = \begin{pmatrix} 0.1 \\ 0.2 \end{pmatrix} + \begin{pmatrix} 0.4 \\ 0.2 \end{pmatrix} + \begin{pmatrix} 0.1 \\ 0.1 \end{pmatrix} + \begin{pmatrix} 0.5 \\ 1.0 \end{pmatrix} = \begin{pmatrix} 1.1 \\ 1.5 \end{pmatrix}$$

For x_2:

$$E_2 = e_{t2} + e_{l2} + e_{f2} + e_{v2} = \begin{pmatrix} 0.2 \\ 0.4 \end{pmatrix} + \begin{pmatrix} 0.6 \\ 0.3 \end{pmatrix} + \begin{pmatrix} 0.2 \\ 0.2 \end{pmatrix} + \begin{pmatrix} 0.4 \\ 0.8 \end{pmatrix} = \begin{pmatrix} 1.4 \\ 1.7 \end{pmatrix}$$

We now form our sequence matrix X_{seq}, where each row is an embedded observation:

$$X_{seq} = \begin{pmatrix} 1.1 & 1.5 \\ 1.4 & 1.7 \end{pmatrix}$$

Step 4: The Basic Transformer Block

We pass X_{seq} through a simplified Self-Attention mechanism. We define our Query, Key, and Value weight matrices as:

$$W_Q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad W_K = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad W_V = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Calculate Q, K, and V:

$$Q = X_{seq}W_Q = \begin{pmatrix} 1.1 & 1.5 \\ 1.4 & 1.7 \end{pmatrix}, \quad K = X_{seq}W_K = \begin{pmatrix} 1.1 & 1.5 \\ 1.4 & 1.7 \end{pmatrix}$$

$$V = X_{seq}W_V = \begin{pmatrix} 1.1 & 1.5 \\ 1.4 & 1.7 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1.5 & 1.1 \\ 1.7 & 1.4 \end{pmatrix}$$

4.1 Attention Scores

Calculate the unscaled attention scores $S = QK^T$:

$$S = \begin{pmatrix} 1.1 & 1.5 \\ 1.4 & 1.7 \end{pmatrix} \begin{pmatrix} 1.1 & 1.4 \\ 1.5 & 1.7 \end{pmatrix} = \begin{pmatrix} 3.46 & 4.09 \\ 4.09 & 4.85 \end{pmatrix}$$

We scale by $\sqrt{d_k} = \sqrt{2} \approx 1.41$:

$$S_{scaled} = \frac{1}{1.41} \begin{pmatrix} 3.46 & 4.09 \\ 4.09 & 4.85 \end{pmatrix} = \begin{pmatrix} 2.45 & 2.90 \\ 2.90 & 3.44 \end{pmatrix}$$

4.2 Softmax Activation

Apply the Softmax function across each row to get the attention weights A.

For Row 1: $e^{2.45} \approx 11.59$, $e^{2.90} \approx 18.17$. Sum $= 29.76$.

$$A_{row1} = [11.59/29.76, \quad 18.17/29.76] = [0.39, \quad 0.61]$$

For Row 2: $e^{2.90} \approx 18.17$, $e^{3.44} \approx 31.19$. Sum $= 49.36$.

$$A_{row2} = [18.17/49.36, \quad 31.19/49.36] = [0.37, \quad 0.63]$$

This gives our attention matrix A:

$$A = \begin{pmatrix} 0.39 & 0.61 \\ 0.37 & 0.63 \end{pmatrix}$$

4.3 Hidden Values

Calculate the final hidden values $H = AV$:

$$H = \begin{pmatrix} 0.39 & 0.61 \\ 0.37 & 0.63 \end{pmatrix} \begin{pmatrix} 1.5 & 1.1 \\ 1.7 & 1.4 \end{pmatrix} = \begin{pmatrix} 1.62 & 1.28 \\ 1.63 & 1.29 \end{pmatrix}$$

Step 5: Flattening and Output Layer

To produce a predicted output for each input, we flatten the hidden states H into a single column vector.

$$H_{flat} = \begin{pmatrix} 1.62 \\ 1.28 \\ 1.63 \\ 1.29 \end{pmatrix}$$

We apply a final simple Neural Network output layer. Let the output weight matrix W_{out} be defined as:

$$W_{out} = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

We calculate the final prediction vector $Y = W_{out} H_{flat}$:

$$Y = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1.62 \\ 1.28 \\ 1.63 \\ 1.29 \end{pmatrix} = \begin{pmatrix} 1.62 + 1.63 \\ 1.28 + 1.29 \end{pmatrix} = \begin{pmatrix} 3.25 \\ 2.57 \end{pmatrix}$$

The final vector gives us a predicted value of 3.25 for the first input and 2.57 for the second input, completing the forward pass of our simplified SERT model.